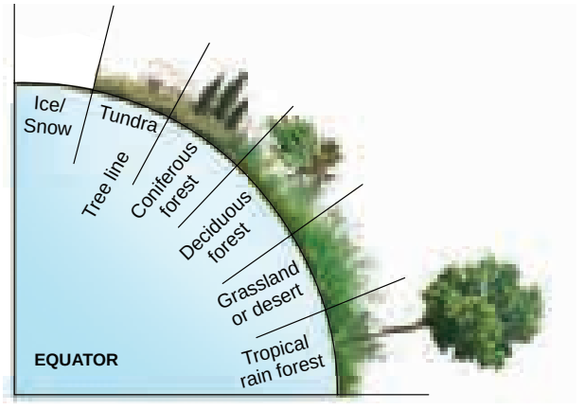


Figure 6.4: Latitudinal zonation of vegetation type

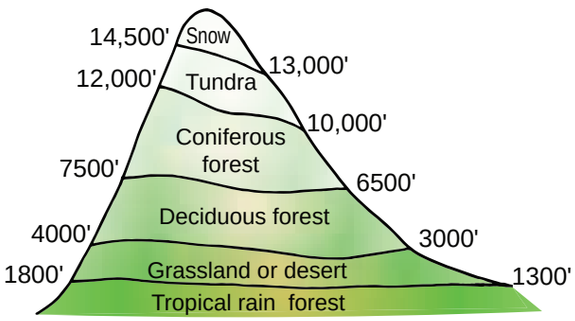


Figure 6.5: Altitudinal zonation of vegetation

Timber line / Tree line : It is an imaginary line in a mountain or higher areas of land that marks the level above which trees do not grow. The altitudinal limit of normal tree growth is about **3000** to **4000m.**

Effects of temperature

The following physiological processes are influenced by temperature:

- Temperature affects the enzymatic action of all the bio-chemical reactions in a plant body.
- It influences CO_2 and O_2 solubility in the biological systems. Increases respiration and stimulates growth of seedlings.
- Low temperature with high humidity can cause spread of diseases in plants.
- The varying temperature with moisture determines the distribution of the vegetation types.

c. Water

Water is one of the most important climatic factors. It affects the vital processes of all living organisms. It is believed that even life had originated only in water during the evolution of Earth. Water covers more than 70% of the earth's surface. In nature, water is available to plants in three ways. They are **atmospheric moisture**, **precipitation** and **soil water**.



Evergreen forests – Found where heavy rainfall occurs throughout the year.

Sclerophyllous forests – Found where heavy rainfall occurs during winter and low rainfall during summer.

The productivity and distribution of plants depend upon the availability of water. Further the quality of water is also important especially for the aquatic organisms. The total amount of water salinity in different water bodies are :i).5% in inland water (Fresh water) ii).30 – 35% in sea water and iii). More than 100% in hypersaline water (**Lagoons**)

Based on the range of tolerance of salinity, organisms are divided into two types.

1. Euryhaline: Organisms which can live in water with wide range of salinity. Examples: Marine algae and marina angiosperms

2. Stenohaline: Organisms which can withstand only small range of salinity. Example: Plants of estuaries.

Terminology		Environmental factor
Stenothermal	Eurythermal	Temperature
Stenohaline	Euryhaline	Salinity
Stenoecious	Euryoecious	Habitat selection (niche)
Stenohydric	Euryhydric	Water
Stenophagic	Euryphagic	Food
Stenobathic	Eurybathic	Depth of water / habitat

Table 6.2: Tolerance of Environmental factor

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